

# Geometry, Entropy, and Operator Learning with Tunable Kernels

Patrick S. Mahon

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## Abstract

We study a two-dial family of kernels

$$k_{\theta,\sigma}(y \mid x) = w_{\theta}(\|x - y\|) g_{\sigma}(y - x),$$

whose product form covers any radial weight  $w_{\theta}$  and centred residual density  $g_{\sigma}$  that satisfy minimal smoothness and moment conditions. To give the theory a concrete footing—and to obtain closed-form bias constants in our finite-sample analysis—we instantiate this family with a data-adaptive *local-linear* construction, but all results hold for the broader class.

For every observable  $f$  we compute the one-step cumulant-generating function (CGF) and show that its quadratic coefficient  $\mathcal{I}_{\theta,\sigma}[f]$  serves simultaneously as (i) a *Fisher load*, (ii)  $\frac{2}{\theta^2}$  times the synthetic Ricci curvature  $\Gamma_2^{(\theta,\sigma)}(f)$ , and (iii)  $-2 \partial_{\sigma}$  of the one-step Shannon entropy. This *Fisher-curvature-entropy bridge* yields a single quadratic objective whose minimiser in the dial plane,  $(\theta^*, \sigma^*)$ —the *little-rose*—optimises curvature bias, entropy production, and operator-learning risk at once. We prove

$$\nabla_{(\theta,\sigma)} \text{KL}(\mu_1 \parallel k_{\theta,\sigma} \# \mu_0) = -\frac{1}{2} \nabla_{(\theta,\sigma)} \mathcal{I}_{\theta,\sigma},$$

so steepest KL descent converges to the little-rose with no hyper-parameters. A data-driven Goldenshluger–Lepski selector recovers  $(\hat{\theta}_{\star}, \hat{\sigma}_{\star})$  from finite samples, and we give non-asymptotic risk bounds for both Perron–Frobenius and Koopman estimates.

The resulting dial matrices  $(\Theta_{ij}, \Sigma_{ij})$  form a directed, doubly-weighted graph, opening avenues toward discrete curvature flows, entropy production on cycles, and multiparameter persistent homology—pointing to a broad research programme at the interface of statistical dynamics, information geometry, and topological data analysis.

## 1 Introduction

### 1.1 Why tune local kernels?

### 1.2 Key contributions

- Closed-form bridge: CGF curvature  $\rightarrow$  Fisher load  $\rightarrow$  curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials  $(\theta, \sigma)$ .
- Finite-sample Goldenshluger–Lepski selector with risk guarantees.

### 1.3 Organisation of the paper

## 2 Rose kernels and dials

### 2.1 Product-form kernels and minimal assumptions (K0–K3)

### 2.2 Local–linear residual construction (canonical example)

### 2.3 Scaled kernels and notation for the rest of the paper

## 3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score  $L^2$ , second-moment scaling.

### 3.1 CGF $\Rightarrow$ Fisher $\Rightarrow$ Landau

### 3.2 Fisher–curvature–entropy bridge

### 3.3 Introducing the dial plane

(K3)  $C^1$  regularity in  $(\theta, \sigma)$

- 3.4 KL gradient and the *little-rose* optimiser
- 3.5 Why it matters (P1–P5 summary)
- 4 Finite-sample dial selection
  - 4.1 Goldenshluger–Lepski selector
  - 4.2 Risk bounds for PF and Koopman estimates
  - 4.3 Illustrative experiment
- 5 Discussion and Outlook
  - 5.1 What we achieved
  - 5.2 Dial-graph viewpoint and future TDA work
  - 5.3 Other open directions (entropy production, curvature flow, etc.)
- A Proof details for Sec. 3
- B GL selector derivations
- C Additional experiments
- D Dial-graph topology (exploratory)